A Republic, If You Can Keep It: Breakdown and Erosion in Modern Democracies

Michael K. Miller, George Washington University

What leads elites to break their democratic commitments? How do the partisan loyalties and democratic values of citizens influence these choices? This article formally analyzes incumbent abuse in democracies, the most common modern initiator of democratic breakdown. If an incumbent attempts to consolidate power, the opposition can try to rally protest but can also cry wolf without incumbent abuse. In a democratic equilibrium, incumbents avoid abuse, the opposition calls for protest if and only if abuse occurs, and citizens protest in sufficient numbers to deter elites. Results show that non-winner-take-all institutions, high media accuracy, low partisanship, and citizen support for democracy typically promote democratic survival. However, under some conditions, higher democracy support can threaten democracy. Model extensions allow incumbents to pursue democratic erosion instead of full breakdown, promote "fake news," and strategically inflame partisanship. The article explores several implications for democratic survival and incumbent strategies for subverting it.

Democracy remains a fragile institution. From 1975 to 2016, 43 democracies broke down, in addition to serious democratic erosion in countries like Hungary, Poland, Nicaragua, and Turkey.[1] Increasingly, democratic breakdowns are initiated not by military coups but by elected leaders broadening their power (Bermeo 2016; Maeda 2010; Svolik 2015). In some cases, like Peru under Alberto Fujimori or Italy under Benito Mussolini, this power grab occurred suddenly. In other cases, such as modern Hungary or Hugo Chavez's Venezuela, democracy suffers a slow decline from the steady expansion of incumbent power.

This article applies a simple formal model to understand these incumbent-led forms of democratic breakdown. In the main model, the incumbent can attempt to consolidate power and render the state autocratic. In an extension, the incumbent has a further option of democratic erosion, a more limited power grab. Either action can be stopped if and only if the opposition mobilizes citizens in protest. In a democratic equilibrium, the incumbent obeys the law and the opposition challenges if and only if a violation occurs.

The goal is to outline the conditions for a "self-enforcing democracy" in which the actions of incumbents, opposition leaders, and citizens mutually reinforce the democratic rules of the game (Przeworski 1991, 2005). Thus, election winners must choose to not abuse their power, election losers must choose to peacefully compete again in the future, and citizens must monitor leaders and police democratic behavior (Graham, Miller, and Strøm 2017; Przeworski 1991).

There are four main strategic reasons the democratic equilibrium can fall apart: temptation to power, co-optation, uncertainty, and partisanship. First, incumbents will abandon democracy if expected protest fails to constrain them. Second, incumbents can offer bribes or intimidate the opposition to accept the power grab. If this advantages both sides, no challenge occurs. Third, citizens do not always observe or know how to interpret their leaders' actions. Although the opposition can call out democratic violations, it can also gain from "crying wolf." Thus, citizens must rely on third-party signaling to evaluate incumbent behavior. Fourth, citizens balance partisan motives against preferences for democracy. In some cases, partisanship is sufficiently high that citizens refuse to protest even if certain of violations. Unlike in many models of protest, this allows for citizens to differ in their preferences for leader change.

Michael K. Miller (mkm2@gwu.edu) is an associate professor of political science and international affairs at George Washington University, Washington, DC 20052.

Data and supporting materials necessary to reproduce the numerical results in the article are available in the *JOP* Dataverse (https://dataverse.harvard.edu/dataverse/jop). An online appendix with supplementary material is available at https://doi.org/10.1086/709146.

1. The breakdown figures use the Polity measure with a threshold of six (Marshall and Jaggers 2017). Using the Boix, Miller, and Rosato (2013) extension yields 37 breakdowns from 1975 to 2015.

The Journal of Politics, volume 83, number 1. Published online December 11, 2020. https://doi.org/10.1086/709146
© 2020 by the Southern Political Science Association. All rights reserved. 0022-3816/2021/8301-0014$10.00

Equilibrium results point to several aspects of power relations and society as influencing whether democracy remains stable. A novel feature of the model is that it does not assume winner-take-all institutions in which electoral winners get all benefits from office. This more realistic assumption is critical to the democratic equilibrium, as democratic survival hinges on achieving a balance between elite actors' benefits from office and their chances at subverting democracy. Offer incumbents too little power, and they will seek to expand it; offer oppositions too little power, and they will resort to false alarms to provoke protest. Thus, flexible and balanced institutions (epitomized by constrained governments, proportional elections, and some forms of power sharing) are conducive to democracy. Democracy is also generally bolstered by high media accuracy, low partisanship, and citizen support for democracy.

The model also has several counterintuitive findings. Surprisingly, citizens' democratic values are never sufficient to secure a democratic equilibrium. In fact, rising democratic values can potentially push a state out of stable democracy when information quality is low. Results also show that higher values for future payoffs (i.e., higher discount factors) are generally bad for democracy and that the opposition's potential to cry wolf is an important neglected element of democratic stability.

Several model extensions explore incumbent techniques for subverting democracy. A key extension allows incumbents to more subtly erode democracy by grabbing only a share of power, which narrows the conditions for stable democracy. The model predicts that incumbents will pursue democratic erosion over full breakdown at middle levels of political strength and low future prospects in democracy. Further, I consider incumbent support for "fake news" and inflamed partisanship, dynamic learning about citizen support for democracy, elite values for democracy, and an incumbent's initial choice of institutions. The article provides a strategic picture of how modern democracies break down, with lessons for how power-hungry incumbents can be constrained.

DEMOCRATIC BREAKDOWN AND EROSION

Key features of modern democratic decline

I begin by outlining several essential elements of modern democratic breakdown and erosion, which are in turn incorporated into this article's model.

Incumbent threats. Democratically elected leaders have become an increasingly common source of democratic breakdown (Bermeo 2016; Maeda 2010; Svolik 2015). According to Maeda (2010), elected leaders accounted for 26% of breakdowns from 1950 to 1991, compared to 61% by external actors (mainly military coups). From 1991 to 2016, in contrast, elected leaders accounted for 56% of breakdowns and external actors only 33% (see fig. 1).[2] This article focuses on incumbent-led breakdowns, with the military assumed not to be the central threat. This is of course critical to understanding motives for breakdown, as the key initiating actor has an alternative option to continue competing within a democracy.

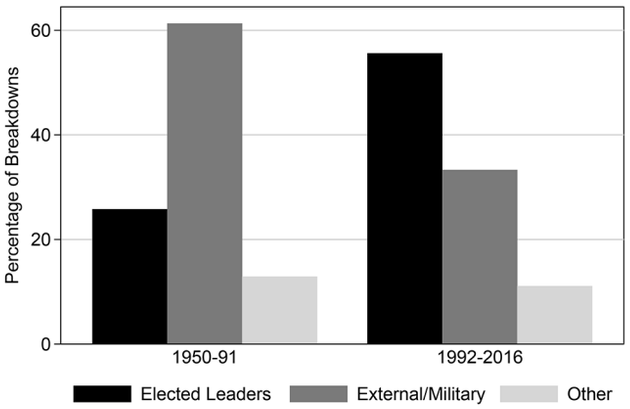

Figure 1. Percentage of democratic breakdowns initiated by democratically elected leaders, external actors (mostly military), and others (coded by Maeda 2010 and me). This is broken down by two periods, with the rise of leader-initiated breakdowns over time clear.

A focus on incumbents is also needed to understand democratic erosion, which is necessarily masterminded by elected leaders and parties. Although harder to define and measure, erosion concerns the slow degradation of democratic norms and institutions to consolidate the incumbent party's power (Bermeo 2016; Coppedge 2017; Huq and Ginsburg 2018; Waldner and Lust 2018). At first, this typically involves nominally constitutional actions that violate the spirit of democratic competition, such as implementing advantageous electoral rules, aggrandizing executive power versus the judiciary and other nonpartisan bodies, and constraining the media and civil society. Coppedge (2017) further distinguishes erosion paths that initially target civil liberties (typified by modern Hungary) and those that erode horizontal accountability, concentrating power in the executive (typified by Chavez's Venezuela). Either path serves to lock in incumbent power for long periods.

Coppedge (2017) identifies 28 episodes of democratic erosion between 1930 and 2015.[3] Following six cases in the early 1930s, there were none until the 1970s, indicating the relative

2. The post-1991 data use Maeda's coding until 2004 (when her data end), plus my coding of 19 post-2004 breakdowns recorded in either Boix et al. (2013) or Marshall and Jaggers (2017), using a threshold of six. See the appendix (available online) for a list.

3. Democratic erosion is defined as a large decline (≥0.09) on the V-Dem liberal democracy score over at least five years in a country that had sustained

popularity of coups. Half the cases (14 of 28) occurred since 1999, including Russia, Turkey, Ecuador, Mexico, and Ukraine. The predominance of erosion in recent history is clear, even without counting ongoing cases like Hungary, Poland, the United States, and the Philippines.

Democratic opposition. Even if they do not initiate democratic breakdowns, opposition elites are important players in their outcomes. Opposition parties are often the most proactive in bringing attention to incumbent abuses of power. Further, citizens face coordination problems when protesting undemocratic actions that can be solved by opposition leaders.

There are two difficulties with the opposition's role, which requires thinking of it as a strategic actor. First, the opposition can be bought off by incumbents (either through payoffs for acquiescence or threats for challenges), leaving the citizenry leaderless and ineffective. Many breakdowns have occurred without significant organized opposition from elites, such as Peru in 1992. This partly explains why several recent cases of democratic erosion—such as Turkey, Hungary, Poland, and Venezuela—were preceded by collapses of traditional parties and party systems, leaving the opposition weak and inexperienced (Seawright 2012).

Second, the opposition can falsely claim violations of democracy and rally protest for political advantage.[4] One example of this "crying wolf" is the phenomenon of presidential impeachments for dubious political reasons, which has been especially common in Latin America (Pérez-Liñnán 2007). Paraguay's Fernando Lugo was removed in 2012 for "blatantly political motives," according to Marsteintredet, Llanos, and Nolte (2013, 116). Other recent cases include Dilma Rousseff's 2016 impeachment in Brazil and Pedro Kuczynski's 2018 resignation (under impeachment threat) in Peru, both arising from criminal accusations that many observers regarded as inflated. Writing in *Federalist* 65 on impeachment, Alexander Hamilton warned of such political maneuvers: "the greatest danger [is] that the decision will be regulated more by the comparative strength of parties than by the real demonstrations of innocence or guilt."

Dangers from opposition behavior are rarely treated as central to models of democratic breakdown, although they are perhaps more familiar from the concept of a "loyal opposition." Yet if opposition leaders can too easily overturn democratic outcomes by claiming violations, democracy is compromised. In turn, this possibility can be a useful rhetorical foil for embattled incumbents and a motivation to preemptively grab power. For instance, Bermeo (2016, 15) argues that Ecuador's history of removing presidents from office early motivated Rafael Correa's power-expanding moves.

Relevance of citizens. Citizens are also critical actors in incumbent-led breakdowns and erosion. The popularity of parties determines their prospects in democracy and their chances of consolidating power undemocratically. As in democratic transitions, citizens are frequently mobilized both in support and opposition to democratic breakdowns, with protests acting either as direct pressure on leaders or as support for institutional processes.

Two motives for citizen protest are worth highlighting: defending democracy and partisanship.[5] First, many citizens have an inherent preference for democratic behavior and will vote, protest, or even commit violence in its defense. Large prodemocracy protest movements have frequently prevented (or stalled) erosion attempts by incumbents, such as in the Philippines in 2001, South Korea in 2016–17, and Poland in 2017 in opposition to judicial reforms (Beissinger 2013; Brancati 2016).

Second, partisanship can overwhelm citizens' democratic commitments, as well as influence how they filter information. This provides popular incumbents room to manipulate rules in their favor. In a survey experiment, Ahlquist et al. (2018) find that giving Hungarians information about "antidemocratic electoral reforms" by the incumbent Fidesz party had little effect on their approval of the reforms. Rather, partisanship was the dominant motivator, with Fidesz supporters regarding the reforms as democratic and legitimate. As a result, "Where strong partisanship binds a sufficiently large minority to the hegemonic party, voters will not throw out the hegemony-seeking incumbent even in fully democratic elections" (918).

Contentious opposition to the incumbent can also be driven by partisanship. Beissinger (2013) shows that seemingly prodemocratic movements like Ukraine's 2004 Orange Revolution can in fact be motivated by motley antiregime grievances and cultural divides. Thailand under Thaksin Shinawatra epitomizes a case with a highly partisan protest movement that organizes around incumbent abuses but displays weak democratic commitments. Such partisan environments present dangerous opportunities for oppositions to rally protest regardless

a minimal democracy level (≥0.5) for at least eight years, was not foreign occupied, and still maintained contested elections.

4. See Magaloni (2010) for a similar application to autocracy, in which opposition parties can falsely allege electoral fraud. Kosterina (2017) shows that ambitious elites' claims against a dictator are less credible because they may be personally advantageous.

5. Similarly, Magaloni (2010) and Svolik (2017) model citizens as trading off disutility from electoral fraud with party preferences.

of incumbent actions. In fact, partisanship can be so intense that it excuses even outright coups. Many Chileans supported the 1974 coup ousting Salvador Allende: "The society had become extremely polarized, and political and social hatred permeated the entire social fabric. . . . A significant part of the population saw the military coup as a solution—a return to order" (Bitar and Lowenthal 2015, 51).

In sum, citizens trade off democratic values and partisan preferences, with heterogeneity in how they are balanced and who citizens support. Further, partisanship need not be understood narrowly as loyalty to a political ideology. In many contexts, political support stems from ethnic, religious, and linguistic identities. In turn, these are often the most politically polarized environments, especially with an absence of cross-cutting cleavages (Dahl 1971).

Uncertainty. Finally, there has been a rise in various types of political takeovers that are unclear in their antidemocratic character. Both democratic erosion and partisan impeachments have uncertain democratic validity, particularly when judged through a partisan lens. As Bermeo (2016, 16) notes, "Contemporary forms of democratic backsliding are most ambiguous and most difficult when they marshal broad popular support—and they often do." Bermeo also points to the rise of "promissory coups" that "frame the ouster of an elected government as a defense of democratic legality" and promise a quick return to elections (8). Fiji, Madagascar, and Honduras recently featured coups sold as stopping elected executives from consolidating power, leaving it unclear which is the democratic side.

We can distinguish two broad types of citizen uncertainty.[6] First, there are unknown actions taken by an incumbent that affect democratic outcomes, such as electoral fraud, bribes to government figures, or covert repression. Second, there are known actions that have unclear democratic valence, such as executive orders, regulations of media outlets, and electoral rule changes. When the opposition brings attention to these events, citizens often cannot tell whether they are serious revelations or posturing for partisan gain.

The idea that democratic citizens can make mistakes in their defense of liberty has a long tradition. Alluding to the XYZ Affair and the Quasi-War with France, Thomas Jefferson wrote in a 1799 letter that, while Americans are genuine republicans, "their virtuous feelings have been played on by some fact with more fiction; they have been the dupes of artful maneuvers, and made for a moment to be willing instruments in forging chains for themselves."[7] To capture the "artful maneuvers" of modern leaders, I model antidemocratic actions as uncertain but the object of informative signaling from the media and civil society.

Empirical and formal models of democratic survival

Despite a large empirical literature on democratic survival, little of it has specifically studied incumbent-led breakdowns. Rare exceptions include Maeda (2010) and Svolik (2015). It therefore remains theoretically unclear whether the dominant empirical approaches to democratic survival—focusing on elite culture (Dahl 1971; Higley and Burton 2006; Mainwaring and Pérez-Liñnán 2013), institutions (Graham et al. 2017; Lijphart 2012; Linz and Stepan 1996), and economic structure (Dahl 1971; Houle 2009)—straightforwardly apply to modern breakdowns. This gap is even greater for democratic decline short of breakdown.[8] Although the concept of "democratic erosion" has been around for several years (e.g., Schedler 1998), there has been little empirical or theoretical work predicting it or contrasting it with full breakdown, as several scholars have recently noted (Bermeo 2016; Waldner and Lust 2018).

Several formal models analyze democratic stability in the face of incumbent threats.[9] The dominant approach is to explain how democracy can be "self-enforcing" (Przeworski 1991, 2005) when elite rivals or citizens react to incumbent abuses. Weingast (1997) examines how citizens can coordinate to deter a ruler's abuse of power, modeled essentially as a repeated Prisoner's Dilemma. Fearon (2011) similarly models incumbent acceptance of democracy when under threat of popular revolt. However, he includes uncertainty about the incumbent's abuse of power, in turn focusing on how elections can signal citizen discontent and coordinate rebellion. Przeworski (2005) and Przeworski, Rivero, and Xi (2015) consider two political parties who can rebel after an election and choice of policy. Democracy hinges on the likely success of each party in armed combat. Chernykh and Svolik (2015) analyze an opposition that chooses to protest fraud on the basis of signals of its popularity, including from a nonstrategic electoral commission. Svolik (2017) models an incumbent choosing a level of electoral fraud. A fraction of voters dislike fraud but have partisan values that may outweigh this

6. I thank Jennifer Gandhi for this point.

7. "From Thomas Jefferson to Thomas Lomax, 12 March 1799," Founders Online, National Archives, https://founders.archives.gov/documents/Jefferson/01-31-02-0056.

8. For a critical review of how well the existing literature explains "democratic backsliding," see Waldner and Lust (2018).

9. Other models of democratic survival focus on threats from military coups or revolutions (e.g., Acemoglu and Robinson 2006; Bohlken 2010).

concern. As a result, political polarization can incentivize incumbents to commit fraud.

By embracing several of these elements, this article's formal model builds on prior work in a few ways. First, it includes the strategic interaction of an incumbent, an opposition, and citizens, whereas most existing models include only one of the latter two as checks on the incumbent. Second, like Svolik (2017), it models citizens as having both partisan interests and concerns about democracy, while also adding, as in Fearon (2011), uncertainty about the incumbent's actions. The payoff is to endogenize citizen activism, with resulting comparative statics that affect survival. Third, the model encompasses a range of institutions under which democracy may or may not survive, rather than assuming winner-take-all. Finally, an extension directly compares the incumbent's motives for a full takeover of power versus democratic erosion, which has been largely unexplored in both formal and empirical work.

MODEL OF DEMOCRATIC SURVIVAL

I begin with a base model with a single round of strategic play and one type of incumbent power grab. As discussed below, a repeated model changes little in the analysis. After exploring the model's implications, I allow for democratic erosion, which only partly expands incumbent power, followed by several other model extensions.

Base model

The model involves three sets of actors: an incumbent ruler, an opposition leader, and an infinite mass of citizens. Each citizen favors either the incumbent or the opposition and may also have a preference for democratic behavior.

Incumbent and opposition. The model begins in democracy and will continue in this state unless the incumbent consolidates power. In the first round, the incumbent is allocated a fraction β of the benefits of office, which total W. This can reflect both rents and policy influence. The opposition gets the remainder, for a payoff of $(1-\beta)W$. In many models, elections are assumed to be winner-take-all (i.e., $\beta = 1$). Allowing β to vary encompasses a wider range of institutional setups, such as proportional representation and power-sharing agreements that formally divide the powers and benefits of office. To explore the most demanding scenario for democracy, I make the common assumption that elites are power seeking and do not inherently value democracy, although a model extension relaxes this.

In the game's first move, the incumbent decides to *violate or not violate* the law to consolidate power. This represents an unconstitutional power grab, which may be an outright self-coup or a shift to electoral authoritarianism with nominal competition. This power grab succeeds unless the opposition rallies a sufficient protest in response. If successful, the country permanently shifts to autocracy, in which the incumbent accrues the benefits of office W indefinitely. If the incumbent violates, he can also offer a transfer $g \geq 0$ to the opposition to not rally protest in response.[10] This represents a co-optation attempt through a one-time payoff, or equivalently, a level of repression applied to oppositions that challenge.

Next, the opposition leader observes whether a violation occurred, as well as the co-optation offer. This is in contrast to citizens, who are less connected to government and must use signals to infer when a violation occurs. The opposition then decides to *challenge or not challenge*, regardless of whether a violation has occurred. Thus, the opposition can "cry wolf" for personal benefit. If she does not challenge after a violation, she gains g but gets 0 payoff within autocracy. If she does not challenge after no violation, the game moves to the next round as a democracy. Both elite players apply a discount $\delta < 1$ to each round.

The expected continuation payoffs for democracy are not assumed to be βW and $(1-\beta)W$ each round. This is unrealistic, as politicians frequently make decisions in anticipation of elections shifting power. Thus, future expectations from democracy are equivalent to $\beta' W$ and $(1-\beta')W$ each round.[11] The fraction β' may be higher or lower than β. This can flexibly incorporate expectations about elections, demographic changes, and so on.

Citizen protest. If the opposition challenges, each citizen opts to *protest or not protest*.[12] I assume that opposition leadership serves a critical coordinating role without which citizens will fail to protest, parallel to the role played by elections in Fearon (2011) and recessions in Acemoglu and Robinson (2006).[13] Once mobilized, citizens will protest depending on personal motivations.

Citizens balance two motives: partisan preference and a desire to protect democracy. A fraction α (with $1/2 \leq \alpha < 1$) of the population supports the incumbent, while a fraction

10. Throughout the article, "he" refers to the incumbent and "she" refers to the opposition.

11. This does not assume that the politicians get these payoffs every round, only that the expected sum is equal to this payoff stream. Thus, the expected continuation payoff for democracy is $\beta' W/(1-\delta)$ for the incumbent and $(1-\beta')W/(1-\delta)$ for the opposition.

12. Protest need not be understood narrowly as marching in the streets but can include activism in support of institutional processes aimed at the incumbent.

13. Alternatively, if citizens can protest without opposition challenge, and we assume there are no opposition voters who protest if certain of no violation or protest fails if only such citizens protest, then the democratic equilibrium condition is identical.

$1 - \alpha$ supports the opposition. Each citizen gets a direct payoff v_O or v_I from supporting his or her favored politician. Thus, an opposition partisan gets v_O from protesting and an incumbent partisan loses v_I from protesting. Unlike in many protest models, citizens therefore differ on whether they want leader change, which is a more realistic framework in a democratic context. In addition, each citizen i has an individual democracy value $d_i \geq 0$, which is drawn from probability distributions with cumulative functions $F_O(d_i)$ and $F_I(d_i)$ for opposition and incumbent supporters, respectively. Citizens gain d_i from choosing the "correct" democratic action, that is, protesting after a violation and not protesting after no violation.[14]

In the model, I assume citizens are intrinsically motivated by the act of protesting.[15] Of course, citizens also care about the protest's outcome, but adding a direct payoff from the outcome does not affect their choice to protest since each citizen has a vanishingly small effect on success. In the global game framework, a popular approach to modeling protest, protesters instead get a payoff if they protest *and* the protest succeeds. In turn, individuals' choices to protest depend on their beliefs about others' choices, generating a coordination dynamic. In the appendix, I show that the equilibrium is substantively identical with such a payoff interaction and the resulting coordination. I focus on the current version for clarity and simplicity.

Because citizens do not directly observe violations, they each have a belief as to whether a violation occurred. For now, I assume that in any equilibrium in which challenges never happen, the off-the-equilibrium-path belief σ of citizens after observing a challenge is initially 1/2 that a violation occurred. The choice of 1/2 is primarily for analytical simplicity, with the equilibrium substantively unchanged for other σ (see the appendix).[16] Following a challenge, each citizen gets an independent signal s_i of whether a violation occurred, with a fraction $q > 1/2$ receiving the correct signal. This can correspond to media reporting, word of mouth, or political messaging from both foreign and domestic actors.[17]

The probability a protest is successful is equal to the fraction of citizens who protest. After a successful protest, the incumbent is ousted and permanently removed from politics (with 0 payoff indefinitely). The opposition gains W for that round but reverts back to $(1 - \beta')W$ in expectation in following rounds as they return to regular democratic competition. If the protest fails and no violation occurred, the roles are reversed—the opposition is permanently removed from politics and the incumbent gains W that round and $\beta' W$ in expectation afterward. If the protest fails and a violation occurred, the country shifts to autocracy.

Summary. The order of play is as follows:

1. The game begins in democracy. The incumbent decides to violate or not violate the law to grab power. If he violates, he chooses a transfer $g \geq 0$ for the opposition to not contest.
2. The opposition leader observes whether or not a violation occurred. In either case, the opposition leader decides to challenge or not challenge. If she does not challenge after a violation, she gets g and the state becomes autocratic.
3. If the opposition challenges, each citizen decides to protest or not protest. Citizens do not directly observe whether a violation occurred. If challenges are off the equilibrium path, each citizen i has an initial belief $\sigma = 1/2$ and gets a signal $s_i \in \{\text{violation, no violation}\}$ (with accuracy q).
4. The likelihood of successful protest is equal to the fraction of citizens who protest. If the protest is successful, the incumbent is ousted and the state remains democratic. If the protest is unsuccessful and a violation occurred, the state becomes autocratic.

It is worth noting that this departs from a typical signaling model, in which one actor has multiple types and others infer their type from signals. Here, all politicians are self-interested, and citizens' uncertainty is over politicians' *hidden actions*.[18] In particular, when citizens see something that should not occur in equilibrium, they are unsure who to blame.

The aim is to solve for a *democratic equilibrium*, defined as an equilibrium in which the incumbent does not violate and the opposition challenges if and only if a violation occurs. Note that this implies challenges occur off the equilibrium path. If citizens see a challenge, they are therefore uncertain whether the incumbent violated or the opposition is crying wolf. The solution concept is *perfect Bayesian equilibrium*. I first consider a one-shot model with strategic play only in the first round, and then I discuss the repeated game in which steps 1–4 repeat as long as democracy survives.

14. Adding a cost to protest has little effect on the equilibrium.

15. For evidence that protesters are intrinsically motivated and do not protest only when they expect others to protest, see Cantoni et al. (2019) and Enikolopov et al. (2017).

16. Setting $\sigma = 1/2$ is also sensible given that there are two politicians with opposing claims.

17. As discussed, many democratic violations are uncertain to citizens, including many coups (Bermeo 2016). However, if a violation is relatively clear, q can be assumed to be close to 1.

18. As a result, separating and pooling concepts do not apply in the usual sense, so I avoid these terms. Further, most standard equilibrium refinements are inapplicable.

Analysis

Citizen protest. We now calculate the fraction of citizens who protest and thus the likelihood of successful protest, depending on whether a violation has occurred. This assumes that the opposition has challenged, as otherwise no protest occurs. Each citizen has an initial belief of 1/2 that a violation occurred before the signal s_i and a q belief corresponding to the signal after.[19] We need to consider four conditions for citizens to protest, based on partisanship and signal.

If incumbent supporters receive a violation signal, they will protest if

$$qd_i > (1-q)d_i + v_I \quad \Rightarrow \quad d_i > \frac{v_I}{2q-1}.$$

If opposition supporters receive a no-violation signal, they will protest if

$$(1-q)d_i + v_O > qd_i \quad \Rightarrow \quad d_i < \frac{v_O}{2q-1}.$$

If opposition supporters receive a violation signal, they will always protest since this maximizes both partisan and democratic preferences. Similarly, if incumbent supporters receive a no-violation signal, they will never protest.

These conditions allow us to calculate the fraction of citizens who protest following violation or no violation. Recall that $F_I(d_i)$ is the cumulative distribution of d_i values for incumbent supporters, a fraction α supports the incumbent, and a fraction q gets the correct signal.

After a violation, the fraction that protests and thus the chance of protest success p_V is

$$\begin{aligned} p_V \equiv \alpha q\left[1 - F_I\left(\frac{v_I}{2q-1}\right)\right] + (1-\alpha)q \\ + (1-\alpha)(1-q)F_O\left(\frac{v_O}{2q-1}\right). \end{aligned} \tag{1}$$

These terms corresponds to incumbent supporters who get the correct signal and have d_i values exceeding $v_I/(2q-1)$, all opposition supporters who get the correct signal, and opposition supporters who get the wrong signal and have d_i values below $v_O/(2q-1)$.[20]

After no violation, the likelihood of successful protest p_{NV} is

$$\begin{aligned} p_{NV} \equiv \alpha(1-q)\left[1 - F_I\left(\frac{v_I}{2q-1}\right)\right] \\ + (1-\alpha)(1-q) + (1-\alpha)qF_O\left(\frac{v_O}{2q-1}\right). \end{aligned} \tag{2}$$

It is straightforward to show that $0 \le p_{NV} \le p_V \le 1$, meaning that violation (weakly) increases the chance of successful protest.[21] The size of this increase, an important quantity for democratic stability, is as follows:

$$\begin{aligned} p_V - p_{NV} = (2q-1)\Big\{\alpha\Big[1 - F_I\Big(\frac{v_I}{2q-1}\Big)\Big] \\ + (1-\alpha)\Big[1 - F_O\Big(\frac{v_O}{2q-1}\Big)\Big]\Big\}. \end{aligned} \tag{3}$$

To interpret this, $1 - F_I[v_I/(2q-1)]$ is the fraction of citizens with democratic values large enough to overwhelm their partisan attachments to the incumbent when they get the correct signal. Thus, the increase's size is a product of information quality (q) and the total fraction of citizens with sufficiently high democratic values.

When will the opposition challenge? The opposition leader can challenge whether or not a violation occurs. In either case, she will anticipate p_V and p_{NV}, as well as the cooptation offer g (considered fixed for now).

After a violation, the opposition will challenge if

$$p_V\left[W + \frac{\delta(1-\beta')W}{1-\delta}\right] + (1-p_V)(0) \ge g. \tag{4}$$

After no violation, the opposition will challenge (cry wolf) if

$$\begin{aligned} p_{NV}\left[W + \frac{\delta(1-\beta')W}{1-\delta}\right] + (1-p_{NV})(0) \\ \ge (1-\beta)W + \frac{\delta(1-\beta')W}{1-\delta}. \end{aligned} \tag{5}$$

A bit of algebra shows that equation (5) is equivalent to

$$p_{NV} \ge 1 - \frac{\beta(1-\delta)}{1-\delta\beta'}. \tag{6}$$

When will the incumbent violate? The incumbent chooses whether to violate (as well as a cooptation offer g). We need to consider two conditions. The first is whether the incumbent favors not violating versus violating and choosing the minimal g to get opposition acquiescence. The second is whether the incumbent favors not violating versus violating, setting $g = 0$, and risking protest. No other choice of g would be rational.

19. The signal has q accuracy, so this corresponds to a simple application of Bayes's Rule.

20. If a cost to protest c is added to the model, then v_I in eq. (1) shifts to $v_I + c$, v_O shifts to $v_O - c$, and similarly for other equations below.

21. Further, $pV > pNV$ unless $F_I[v_I/(2q-1)] = F_O[v_O/(2q-1)] = 1$.

Consider the first possibility in which a sufficient g is offered, an amount calculated in equation (4). The incumbent prefers no violation to violating and offering g if

$$\beta W + \frac{\delta\beta' W}{1-\delta} \geq \frac{W}{1-\delta} - g \tag{7}$$

$$\Rightarrow \beta W + \frac{\delta\beta' W}{1-\delta} \geq \frac{W}{1-\delta} - p_V\left[W + \frac{\delta(1-\beta')W}{1-\delta}\right] \tag{8}$$

$$\Rightarrow p_V \geq \frac{1-\beta+\delta(\beta-\beta')}{1-\delta\beta'} = 1 - \frac{\beta(1-\delta)}{1-\delta\beta'}. \tag{9}$$

Now suppose the opposition will challenge if and only if a violation has occurred. Now the incumbent chooses no violation if

$$\beta W + \frac{\delta\beta' W}{1-\delta} \geq p_V(0) + (1-p_V)\frac{W}{1-\delta}$$

$$\Rightarrow p_V \geq 1 - \beta + \delta(\beta - \beta').$$

This condition is less strict than the last, so the first condition for incumbent behavior binds. It also follows that, after violating, the incumbent always weakly prefers offering sufficient g to deter a challenge (strictly if $\delta, \beta' > 0$).

Democratic equilibrium condition. Putting these conditions together, we can derive the requirements for a democratic equilibrium in which the incumbent does not violate (eq. [9]) and the opposition challenges if and only if the incumbent violates (eqq. [4] and [6]). All proofs are in the appendix.

> **Proposition 1.** In the one-shot game, a democratic equilibrium holds if and only if
>
> $$p_V \geq 1 - \frac{\beta(1-\delta)}{1-\delta\beta'} \geq p_{NV},$$
>
> where p_V and p_{NV} are defined in equations (1) and (2). The equilibrium is unique when the inequalities are strict.

Thus, $p_V - p_{NV}$ captures the space in which β and β' can be set to secure democracy. Comparative statics and the implications of proposition 1 are covered below. For now, note that $1 - [\beta(1-\delta)/(1-\delta\beta')]$ strictly declines with β and β' and can always be set to fall outside the given bounds. Thus, power needs to be allocated appropriately for democratic behavior to result.

In the appendix, I overview the full set of pure-strategy equilibriums in this base game. Omitting knife-edge conditions, the only remaining equilibrium is where the incumbent violates and the opposition accepts a payoff to not challenge. Like the democratic equilibrium, it is unique when its conditions are satisfied strictly. I also allow for different off-the-equilibrium-path beliefs σ. The equilibrium condition is substantively identical to proposition 1, with p_V and p_{NV} replaced by $p_V(\sigma)$ and $p_{NV}(\sigma)$, which both weakly increase with σ. The implication is that democracy can partly be a matter of mutual expectations and not just incentives (Little, Tucker, and LaGatta 2015; Weingast 1997).

In the base model, only the first round involves strategic play. As shown in the appendix, extending this to a repeated game is straightforward and involves little adjustment to the equilibrium, even when allowing the parameters (α, β, etc.) to dynamically change each round. In fact, the condition for a permanent democratic equilibrium is identical to proposition 1 but must now be met in every round.[22] Since the repeated game offers little substantive difference, I keep the notation for the one-shot game in the remainder of the article.

Special cases

The model allows β and β' to encompass a wide range of political institutions. It is instructive to consider how the equilibrium condition changes with some natural specifications.

Winner-take-all. A common critique of majoritarian institutions is that they are winner-take-all, producing power imbalances. If $\beta = 1$, proposition 1 implies that β' must satisfy

$$\frac{\delta - p_V}{\delta(1-p_V)} \leq \beta' \leq \frac{\delta - p_{NV}}{\delta(1-p_{NV})}.$$

If $\beta' = 1$ as well, this can never be satisfied since oppositions permanently cut out of power clearly have an incentive to challenge. Further, the condition fails for all β' if $p_{NV} > \delta$. Thus, it is not necessarily sufficient for the opposition to have a chance to win elections in the future.

Radical equality. Suppose $\beta = \beta' = 1/2$, a condition in which power is equally shared regardless of elections. The condition in proposition 1 now shifts to

$$p_V \geq \frac{1}{2-\delta} \geq p_{NV}.$$

Note that this rapidly becomes impossible to satisfy as δ increases because the incumbent's incentive to grab more than equal power becomes too high.

22. As discussed in the appendix, if we allow this condition to possibly fail in the future, the exact condition is changed but remains substantively similar.

Stable expectations. Suppose $\beta = \beta'$, meaning that power relations are highly stable. The condition in proposition 1 now simplifies to

$$p_V \geq \frac{1-\beta}{1-\delta\beta} \geq p_{NV}.$$

Incumbent disaster. Suppose $\beta' = 0$, meaning the incumbent will be uncompetitive in the future, perhaps because of economic failure or legal punishment. Proposition 1 now shifts to

$$p_V \geq 1 - \beta(1-\delta) \geq p_{NV}.$$

Note this can be satisfied only for very large p_V as δ increases because the incumbent, for obvious reasons, is incentivized to grab power. Thus, strong ex-incumbents may need to be given some share of power ($\beta' > 0$) as a condition of tolerating democratic rule. For instance, the Portuguese junta that accepted democracy in 1976 held an effective veto on legislation until 1982. Chile's outgoing military dictator, Augusto Pinochet, retained control of the military for nearly a decade after democratization.

Single round. Suppose $\delta = 0$, so that both elite players do not value power in the future. The condition in proposition 1 now shifts to

$$p_V \geq 1 - \beta \geq p_{NV}.$$

This is instructive as it shows that β can be adjusted to achieve democratic cooperation even without the shadow of the future, unlike a repeated Prisoner's Dilemma.

IMPLICATIONS

The equilibrium condition in proposition 1 (and parallel condition for the repeated game) has several implications for understanding democratic stability.

Institutions

A key result is that democratic stability requires limits on the current and expected future power of elites. Specifically, β and β' must be moderate for democracy to survive. This is pictured in figure 2, which shows the ranges of β and β' that support a democratic equilibrium. As both trend toward 0, the incumbent is driven to grab power. As both trend toward 1, the opposition is shut out from power and will resort to crying wolf, incentivizing the incumbent to violate in anticipation. In contrast, there is always a middle $\beta = \beta'$ that satisfies the condition in proposition 1:

$$\frac{1-p_V}{1-p_V\delta} \leq \beta = \beta' \leq \frac{1-p_{NV}}{1-p_{NV}\delta}.$$

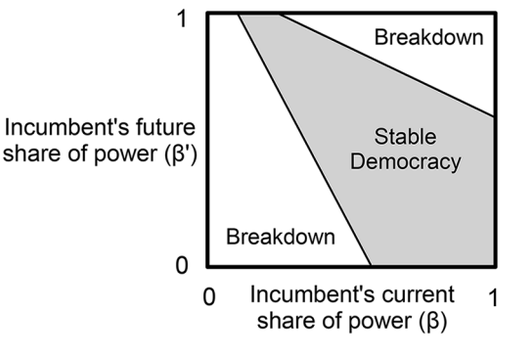

Figure 2. Conditions for a democratic equilibrium, as formulated in proposition 1, based on the incumbent's current share of power (β) and expected future share of power (β'). If both are too high or too low, democracy breaks down. Only when the benefits of office are balanced will democracy be stable. Assumptions: $p_V = 0.95$, $p_{NV} = 0.7$, and $\delta = 0.9$.

This demonstrates the importance of institutions that moderately balance power, such as proportional representation, power-sharing arrangements, divided cabinets, and governmental constraints. Winner-take-all institutions (exemplified by strong presidents, plurality rule, and limited checks and balances) are often too favorable to majorities to make democratic behavior preferable for minority groups (Lijphart 2012). In particular, power should be allocated in line with group capacities to overturn democratic institutions (p_V, p_{NV}), a key premise of postconflict power sharing (Roeder and Rothchild 2005).

Expectations of future power are also critical—all that is needed for democracy to die is for β' to fail the condition at some moment.[23] This helps to explain why democratic failures are so common around elections,[24] as well as why conflicts over term limits have been so destabilizing in countries like Honduras, Zambia, and Burkina Faso. It also provides a mechanism for economic crises' contributions to breakdowns, as incumbents expect poor electoral outcomes, driving β' down. Again, moderate institutions can limit the expected shifts in power.

Institutions and party systems can also affect the incumbent's likelihood of consolidating power. For instance, strong internal constraints on executive power should raise p_V, supporting democracy. In contrast, fractionalized party systems and those following traditional party collapses should find coordination and mobilization against incumbent abuses more difficult (Seawright 2012), lowering p_V and making democracy more vulnerable.

23. As Tadeusz Mazowiecki (Poland's first post-1989 prime minister) put it, stability in Poland required "that everyone had a future in democracy" (Bitar and Lowenthal 2015, 280).

24. Examples include Franco's military revolt after the Spanish Republicans' 1936 election win, the 1967 Greek coup preempting a center-left victory, and the 2006 and 2014 Thai coups.

As empirical support of these implications, Graham et al. (2017) compare how several power-sharing institutions contribute to democratic survival. They find that institutions that constrain incumbent power (such as strong independent judiciaries) are positive for survival under all conditions. Institutions that mandate central government inclusiveness (such as ethnic quotas) are positive for survival following civil wars, a situation typically corresponding to high partisanship. Finally, institutions that divide power territorially (such as federalism) are negative for survival after civil war.[25] More generally, Graham et al. concur with this article's formal results that power-sharing institutions have context-dependent effects and thus should be tailored to each country's political situation.

South Africa's successful installation of democracy also illustrates the model's logic well. The 1993 agreement that ended apartheid included a five-year grand coalition deal between the outgoing National Party and African National Congress, proportional representation, and 34 guarantees regarding the later constitution, including a strong Constitutional Court (Sparks 1996). Reflecting on the agreement, African National Congress leader and future president Thabo Mbeki remarked, "Of course, in elections, one party wins and the other party is defeated. But you can't draft a constitution on the basis of majorities and minorities. Even the minority must get a sense of ownership" (Bitar and Lowenthal 2015, 330).

Partisanship and democratic norms

Levels of partisanship (v_I, v_O) and citizen values for democracy $\{d_i\}$ affect democratic survival by influencing elites' outside options against democracy (captured by p_V, p_{NV}). Since democracy is in equilibrium when a function of institutional power lies between p_V and p_{NV}, factors that raise p_V or lower p_{NV} will support democracy.

To capture these comparative statics formally, I simplify slightly by assuming that both sides have the same level of partisanship (v) and distribution of democratic values. Specifically, I make the following two-part assumption:

Assumption 1.

1. $v = v_I = v_O$.
2. $F(d_i) = F_I(d_i) = F_O(d_i)$, with probability density function $f(d_i)$.

Corollary 1. Given assumption 1, if $f[v/(2q-1)] = 0$, then $dp_V/dv = dp_{NV}/dv = 0$. Otherwise, the following statements hold:

1. $dp_V/dv < 0$.
2. $dp_{NV}/dv < 0$ if $q < \alpha$, $dp_{NV}/dv = 0$ if $q = \alpha$, and $dp_{NV}/dv > 0$ if $q > \alpha$.
3. $d(p_V - p_{NV})/dv < 0$.
4. Globally increasing democratic values d_i (i.e., choosing $F'(d_i)$ that strictly first-order stochastically dominates $F(d_i)$) has the opposite direction of effect as increasing v for all the above comparative statics.

Higher partisanship acts as a direct counter to democratic values and is generally bad for sustaining democracy. Specifically, higher democratic values and lower partisanship always widen $p_V - p_{NV}$ and thus the bounds for the democratic equilibrium. This is sensible as more citizens are willing to protest undemocratic behavior.

In the limit of large $\{d_i\}$ relative to v, p_V converges to q and p_{NV} converges to $1 - q$ as citizens protest if and only they get a violation signal. In turn, the democratic equilibrium then requires $1 - q < \beta(1-\delta)/(1-\delta\beta') < q$. This lays out clearly the dependence on media quality when partisanship is low.

In the limit of large v relative to $\{d_i\}$, both p_V and p_{NV} converge to $1 - \alpha$ as citizens protest if and only if they oppose the incumbent. The democratic equilibrium will hold (through indifference) only if β and β' satisfy a razor-edge condition, namely, $\alpha = \beta(1-\delta)/(1-\delta\beta')$. If $\delta = 0$ (i.e., no future rounds matter), this reduces to $\alpha = \beta$, meaning power and population need to perfectly track each other in highly partisan environments. This reiterates the logic of quota-based power sharing after civil wars, such as Burundi's constitutional mandate that its national assembly be a fixed percentage of Hutu and Tutsi.

Corollary 1 also has counterintuitive implications, as raising $\{d_i\}$ for all citizens or lowering v does not always make democratic survival more likely. In fact, either move can shift out of a democratic equilibrium when media accuracy is low. The reason for this paradoxical result is that increasing democratic values $\{d_i\}$ has opposing effects on two groups when the incumbent does not violate. Incumbent supporters who get an incorrect signal are more likely to protest, but opposition supporters who get a correct signal are less likely. When $\alpha > q$, the first effect dominates and p_{NV} increases. This in turn makes it more tempting for the opposition to cry wolf, potentially shifting out of a democratic equilibrium if $1 - [\beta(1-\delta)/(1-\delta\beta')]$ is just above p_{NV}. The

25. Following Graham et al.'s (2017) theory, the negative effect of territorial power sharing can be understood as a product of lower q as different factions retreat to their separate corners and avoid centralized bargaining and information sharing. It may also correspond to higher p_{NV}, as dominant regional powers can mobilize their local populations.

same applies to lowering v. The implication is that citizen values can often contribute positively to democratic survival but cannot guarantee it—the information environment and elite incentives always need to align as well.

Media and information reliability

Information reliability, best captured by media quality, has clearer positive effects on democratic survival. Again, factors that raise p_V and lower p_{NV} are conducive to democracy. The relevant effects of q are summarized as follows:

Corollary 2. Given assumption 1, if $F[v/(2q-1)] = 1$, then $dp_V/dq = dp_{NV}/dq = 0$. Otherwise, the following statements hold:

1. $dp_V/dq > 0$.
2. Increasing q by a sufficient amount strictly lowers p_{NV}.
3. $d(p_V - p_{NV})/dq > 0$.

A sufficiently higher q always makes the democratic equilibrium more likely because it raises p_V and lowers p_{NV}. To be clear, the effect on p_{NV} is not always monotonic—depending on other parameters, there can exist a small range over which p_{NV} increases with q. However, a sufficiently sized increase in q is guaranteed to strictly lower p_{NV}.

The results point to the importance of a high-quality, independent media in transmitting reliable information to citizens. As American Senator Jeff Flake remarked, without "a principled fidelity to truth and to shared facts . . . our democracy will not last."[26] Indeed, a recent Gallup poll found that 84% of Americans see news media as "critical" or "very important" to democracy (Ritter and Jones 2018). If the media is controlled by the state, highly fractured by partisan interests, or untrustworthy, its informational role is likely to fail. In turn, incumbents may have an incentive to attack or degrade news reliability (see below).

Several fragile democracies have seen the opposition claim electoral fraud and try to rally protest against the government, followed by outside parties signaling the election was reasonably fair, facilitating democratic survival. For instance, the opposition New Patriotic Party disputed Ghana's 2012 election, claiming "widespread abuse" and "manipulated" results (BBC 2012). However, the election featured two international monitoring teams, several thousand domestic monitors, and an independent electoral commission that gave positive evaluations of the election. For the ensuing trial, the Supreme Court broadcast the entire 50-day case live on television and radio. Widely watched by Ghanaians, the trial affirmed the results and further signaled the election's legitimacy. Ultimately, there was little popular support for challenging the results, and the opposition relented. This demonstrates the critical importance of having credible third parties to signal information.[27]

The military and external actors

Although not explicitly included in the model, we can easily take into account an outside role for the military and other actors (including foreign actors). Most simply, we can imagine them shifting p_V and p_{NV} up or down, depending on whom they support, which can in turn push the state out of a democratic equilibrium. An interesting feature of this dynamic is that military support may terminate democracy without the military initiating a coup but simply making it clear whom they would support in a conflict. In Madagascar's political crisis in 2009, for example, the army did not initiate the violent overthrow of the president but ultimately sided with the opposition. Had the army's loyalty been more secure, the revolt—led by the capital's mayor—may never have happened.

Alternatively, we might imagine prodemocratic external actors that choose to punish perceived attempts to grab power. Below, an extension allows for elite democratic rewards. However, just like with citizens, this hinges on whether these external actors have accurate information and reliably enforce democratic behavior despite personal benefits from empowering specific parties. Unfortunately, external democratic enforcement has frequently been weak because of uncertainty on how to interpret democratic erosion and geopolitical interests that overwhelm democratic commitments (Bermeo 2016, 15–16; Goldsmith 2008).[28]

Time discounting

Finally, a democratic equilibrium requires that δ be moderate (or at least not too high). As δ approaches 1, the condition in proposition 1 cannot be satisfied unless $p_V = 1$ (which in turn requires $q = 1$). This is surprising, as typically we think of high discount factors as securing cooperative behavior. However, here a high δ makes the consolidation of power too attractive, and the democratic compact

26. Senate speech, January 17, 2018: 164 Cong. Rec. (10) S215.

27. Similarly, Ernesto Zedillo (Mexico's president from 1994 to 2000) credited the signaling power of institutions with avoiding "catastrophic consequences" for democracy when the opposition claimed fraud in the 2006 Mexican election (Bitar and Lowenthal 2015, 191).

28. Regarding democratic erosion, consider the European Union's hesitancy to punish Hungary and Poland for antidemocratic behavior.

breaks down, more in line with credible commitment models of war (Powell 2006).

DEMOCRATIC EROSION

I now explore the widening phenomenon in which incumbents subtly erode democratic institutions rather than grabbing all power in a self-coup. Exemplified by Hungary, Turkey, and Venezuela, this involves manipulating electoral and institutional rules to lock in a large share of power while maintaining the procedural veneer of democracy. When will incumbents opt for erosion over a self-coup?

To examine this formally, assume the incumbent now has two antidemocratic options: *erosion* and *self-coup*. The latter is the same as violation in the earlier model. For erosion, the incumbent instead chooses a fraction of power β^* up to a limit $\bar{\beta} \leq 1$ that he will permanently grab if successful. Assume that erosion is only an option if $\bar{\beta} > (1-\delta)\beta + \delta\beta'$, meaning that $\bar{\beta}$ is an improvement over what the incumbent expects in democracy. If choosing erosion, the incumbent can no longer offer a payoff g given the difficulty of repressing or credibly offering bribes while maintaining a democratic facade.

The opposition can claim erosion, self-coup, or no violation. For simplicity, I assume that citizens form a belief as to whether any violation occurred but are indifferent as to the type of violation. If the opposition claims erosion or self-coup and this occurs off the equilibrium path, citizens have an initial 1/2 belief in violation. Citizens then get an independent signal as to whether the claimed violation occurred, with accuracy q if the opposition claims self-coup. Since erosion is harder to detect, I assume only a $\bar{q} < q$ accuracy if the opposition claims erosion. The protest stage proceeds as before.

This framework incorporates several distinctive features of democratic erosion: it does not consolidate power to the same degree as full autocratic takeover, it is relatively constrained in the techniques of manipulation, and it is more subtle and thus harder to perceive. Incumbents still face the demonstrated trade-off between accepting democracy and abusing their power. Now, they also must decide whether to grab less power with a lower chance of detection.

The following proposition describes equilibrium behavior in a one-shot game, equivalent to proposition 1 with the erosion option added. Equilibrium results for a repeated game are again virtually identical. Let $\bar{p}_V$ and $\bar{p}_{NV}$ be the analogues of p_V and p_{NV} from equations (1) and (2) with $\bar{q}$ in place of q. Note that corollary 2 implies that $\bar{p}_V \leq p_V$.

Proposition 2. In the one-shot game with the democratic erosion option, the proposition 1 condition applies if $\bar{\beta} \leq (1-\delta)\beta + \delta\beta'$. Otherwise, the following equilibrium conditions apply, with the equilibriums unique when satisfied:

- A democratic equilibrium holds if and only if $\bar{p}_V \geq 1 - [\beta(1-\delta)/(1-\delta\beta')] \geq \max\{p_{NV}, \bar{p}_{NV}\}$.
- In any equilibrium in which the incumbent chooses erosion, the incumbent erodes to $\beta^* = \min\{1 - \bar{p}_V(1-\delta\beta'), \bar{\beta}\}$ and the opposition does not challenge. This requires that $\bar{p}_V < 1 - [\beta(1-\delta)/(1-\delta\beta')]$ and $\bar{\beta} \geq 1 - p_V(1-\delta\beta')$.
- Excluding knife-edge conditions, the only remaining pure-strategy equilibrium is when the incumbent chooses self-coup and offers sufficient payoff to avoid a challenge.

The availability of erosion narrows the conditions for democratic survival. The key bounds p_V and p_{NV} within which we can set institutional power constrict to $\bar{p}_V$ and $\max\{p_{NV}, \bar{p}_{NV}\}$. Thus, if erosion is a realistic option, liberal democracy becomes more fragile and institutional design more critical.

In addition, the proposition tells us when regimes will pursue democratic erosion versus self-coups. See the left side of figure 3 for example ranges of β and β' that support democracy, erosion, and self-coups in equilibrium. It is worth first explaining how erosion works formally. The key is that incumbents can costlessly erode up to a specific level of power without provoking challenge, which they will if this level exceeds what they expect in democracy. The level rises as $\bar{p}_V$ falls since challenge becomes less desirable for the opposition. Further, as β and β' decline, the level is more likely to exceed democratic returns. Finally, erosion is preferable to a self-coup when maximal erosion $\bar{\beta}$ is high, p_V is high (deterring coups), and β' is low. The latter holds because lower β' means the incumbent must pay a higher g to deter challenge.

Three significant implications arise from this. First, erosion is more likely when $\bar{\beta}$ is higher, meaning more power can be grabbed while still alerting citizens at the less accurate rate $\bar{q}$. For instance, younger democracies with more flexible interpretations of democratic behavior and weaker constraints on power consolidation should be more vulnerable to erosion.

Second, erosion is more likely at low β', meaning when incumbents have low expectations of future power in democracy. In response, erosion locks in a high share of power. Thus, erosion may be especially attractive to ruling parties that perceive a steady loss of electoral support or predict demographic changes that will challenge their future dominance.

Third, erosion is more likely at middle values of incumbent strength, at least as reflected in the capacity to survive protest. Specifically, erosion is in equilibrium when $\bar{p}_V$ falls

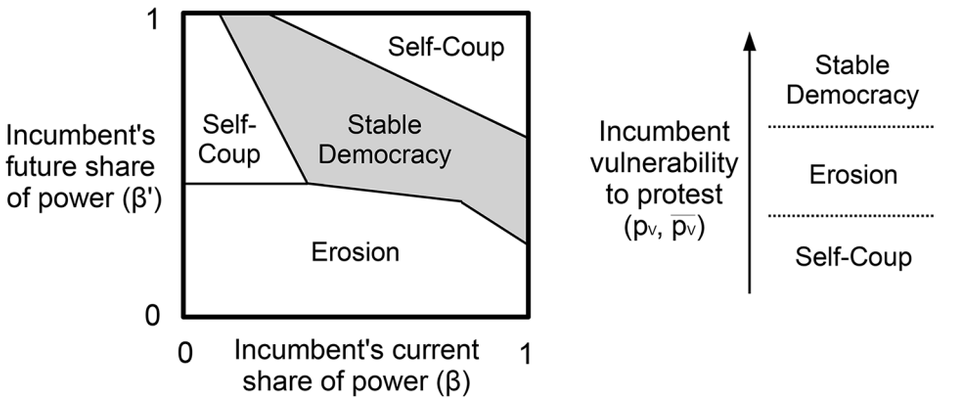

Figure 3. Conditions for a democratic equilibrium when democratic erosion is an option. *Left*, based on the incumbent's current share of power (β) and expected future share of power (β'). If both are too low, the incumbent favors democratic erosion. If both are moderate, a democratic equilibrium can hold. If β' is high, the incumbent is more likely to favor a self-coup. Assumptions: $p_V = 0.95$, $p_{NV} = 0.7$, $\bar{p}_V = 0.85$, $\bar{p}_{NV} = 0.73$, and $\delta = 0.9$. *Right*, equilibrium outcomes as p_V and $\bar{p}_V$ increase, with erosion resulting for middle levels of vulnerability to protest.

below a threshold and p_V lies above a lower threshold. Various factors, such as the fraction of incumbent supporters α and military support, could mutually affect p_V and $\bar{p}_V$. If we imagine them moving together, erosion occurs for a middle range of p_V and $\bar{p}_V$. In comparison, democracy holds for high values and self-coups for low values. See the right side of figure 3 for a representation of this.

In sum, moderately strong incumbents—with sufficient power to manipulate democratic institutions but not enough to easily grab all power—who face uncertain or declining future prospects in democracy are the most likely to pursue erosion. Strong incumbents who can survive challenges and who have too much control in democracy for the opposition to remain loyal are the most likely to launch self-coups. Finally, democracy is stable when incumbents are highly vulnerable to protest if they violate democracy and when the benefits of democracy are balanced between the incumbent and opposition.

OTHER EXTENSIONS

I now consider several more extensions, with further details and proofs in the appendix.

How democratic are the people? Learning about d_i

In the model, it is assumed that the distribution of d_i (which affects p_V and p_{NV}) is known to all players. A simple extension leaves this exact distribution uncertain. An implication of this is that various events can shift the game into a democratic equilibrium if citizens are given a chance to protest. For instance, the first round of a repeated game could involve the incumbent violating, the opposition challenging, and a sufficient number of citizens protesting so that beliefs are updated and a democratic equilibrium holds in all future rounds. This bears resemblance to several people-powered revolutions in which mass protest movements revealed high support for democracy and led to fairly stable democracies, such as Czechoslovakia, South Korea, and Benin. Indeed, Bayer, Bethke, and Lambach (2016) find that democracies survive longer when their transitions feature large nonviolent protest movements.

What if elites value democracy? Adding d_{inc} and d_{opp}

The model deliberately assumes that elites care solely about power. What if they value democracy for either normative or material reasons, including international rewards? Consider adding values $d_{\text{inc}}, d_{\text{opp}} > 0$ for the incumbent and opposition, respectively, for maintaining democracy. Unsurprisingly, these parameters strictly widen the conditions for the democratic equilibrium.[29] However, both values need to be very high to guarantee democracy. Specifically, the proposition 1 condition holds for all p_V, p_{NV} if and only if

$$d_{\text{opp}} \geq \beta W \quad \text{and} \quad d_{\text{inc}} \geq W\left[(1-\beta) + \frac{\delta(1-\beta')}{1-\delta}\right].$$

The final term is the opposition's total expected value of remaining in democracy. Thus, elites' democratic motives generally must exceed their values for power for democratic survival to not depend on citizen behavior.

Do incumbents support "fake news"? Strategically adjusting q

A simple extension allows the incumbent to invest resources to raise or lower q. Depending on the parameters, it may be in the incumbent's interest to raise q so that a democratic equilibrium holds. Alternatively, the incumbent may prefer

29. Higher d_{opp} relaxes the cry-wolf condition for p_{NV}, and both relax the condition for p_V.

to lower q so that a violation is likely to succeed (since p_V declines with lower q). If the cost of shifting down is sufficiently low, then all incumbents will do this as it guarantees consolidation of power at low cost.

Suppose that q can be adjusted as a function of cost c and that an incumbent wants to lower q and in turn p_V, intending to launch a self-coup. Consider the implied function $p_V(c)$, with $p_V > 0, p'_V < 0$, and $p''_V > 0$.[30] Thus, the incumbent maximizes

$$\frac{W}{1-\delta} - p_V(c)\left[W + \frac{\delta(1-\beta')W}{1-\delta}\right] - c.$$

It is simple to prove the incumbent will choose c so that

$$p'_V(c) = -\frac{1-\delta}{W(1-\delta\beta')}$$

(assuming this is satisfied for $c > 0$). If an incumbent erodes instead, then the same condition holds for $\bar{p}_V{}'(c)$ (assuming this is satisfied for $c > 0$ and $\beta^* \leq \bar{\beta}$). Thus, the media will be degraded more at higher W and lower β'.

The implication is that antidemocratic incumbents will expend effort to contribute to a degraded and untrustworthy media environment (low q). This can involve several techniques, including promoting partisan or state media, attacking unfriendly media as "fake news," increasing government secrecy, and censoring or closing down media outlets (Simon 2006; Waldner and Lust 2018). Starting in 2004, Turkey under Erdoğan restricted press coverage through defamation laws, bans on coverage of sensitive issues, and state leverage over media companies (Bermeo 2016, 11). Similar moves occurred in Venezuela, Poland, Russia, and Thailand (Huq and Ginsburg 2018). This flawed informational environment paralyzes citizen activism and makes democratic violations more likely.

A similar analysis allows the incumbent to invest in higher polarization, a common rhetorical strategy of antidemocratic populists. If an incumbent intends to violate democracy, it will always be profitable to inflame partisanship to make violation more likely to succeed. This can include employing incendiary rhetoric, scapegoating opposition groups for perceived harms, exploiting sensitive ethnoreligious divides, and encouraging multisided violence. Alternatively, incumbents can try to reduce popular preferences for democracy by blocking international norms, prioritizing contrary values (such as security or nationalism), or attacking democracy itself (Cooley 2015). Hungary's Prime Minister Viktor Orbán has been forthright on this goal, arguing in a 2018 speech that Western democracy is "shipwrecked" and "the era of liberal democracy is over" (Santora and Bienvenu 2018).

How much power is too much? Choosing β and β'

Finally, suppose the incumbent influences institutional design, such as at the founding of a democracy. Founders often need to choose between proportional or winner-take-all electoral systems, for instance. Specifically, let an incumbent initially choose β and β'. For simplicity, I assume that $\beta = \beta'$. Note that this differs from the democratic erosion model because choosing high β is no longer a violation of democracy.

Incumbents are tempted to raise β since this increases their potential payoff. However, an interesting result is that incumbents may prefer a lower β if this shifts into a democratic equilibrium, although they will always choose the maximum β for which the equilibrium holds. The condition for incumbents to choose this democratic β over $\beta = 1$ is

$$p_V \geq \frac{p_{NV}}{1-\delta p_{NV}} \quad \Leftrightarrow \quad \frac{1}{p_{NV}} - \frac{1}{p_V} \geq \delta.$$

This is derived from comparing the maximal β at which proposition 1 holds to the expected payoff from setting $\beta = 1$ and offering a sufficient g after violating.

Once again, this condition tells us that a high chance of successful protest after violation and a low chance of successful protest if the opposition cries wolf are both critical to democracy. A surprising implication is that the incumbent is more likely to choose democracy at lower δ.

CONCLUSION

This article considered a formal model of democratic survival, with a focus on incumbent takeovers of power. To secure democracy, an opposition must be able to rally the citizenry to deter incumbent violations yet not be tempted to cry wolf for political advantage. Although relatively simple, the model provides an implication-rich framework for understanding democratic stability and how incumbents undermine it.

In the main result, democracy is generally more stable when institutions balance power and benefits of office between election winners and losers, when citizens have reliable information sources on their leaders' behavior, and when citizens do not value partisanship too strongly over democratic behavior. More surprisingly, higher democratic values and lower partisanship can sometimes threaten democracy when information quality is low. In addition, higher elite values for future payoffs typically undermine democracy.

The implications bode poorly for the stability of American democracy. Although there is an ongoing debate on Americans' democratic preferences, there is strong evidence of accelerating polarization and declining trust in news. Despite a

30. This implies c lowers p_V with declining marginal effectiveness, but p_V never reaches 0.

venerated tradition of checks and balances, the United States features several winner-take-all institutions, including an elected president with weakly defined powers, first-past-the-post elections, an increasingly partisan Supreme Court, and an unusual system of party-controlled state electoral administration (Huq and Ginsburg 2018). Whether these institutions face increasing strain or continue to coexist with constrained executive power remains an open question.

Model extensions predict how antidemocratic incumbents will strategically undermine democracy. This may involve democratic erosion, especially when incumbents are moderately strong and can lock in a large advantage at minimal chance of detection. Power-hungry incumbents should try to inflame partisanship, even at the cost of further repelling opponents, and reduce the reliability and popular trust of national media. As it happens, expanding executive power, using populist rhetoric to target minority groups and intensify partisanship, and attacking the media are among the key warning signs of democratic backsliding (Bermeo 2016; Huq and Ginsburg 2018; Schedler 1998; Waldner and Lust 2018).

Scholars should continue to investigate (both empirically and formally) how modern democratic breakdown and erosion operate. Evidence shows that would-be autocrats learn how to subvert democracy from one another's examples (Bermeo 2016; Cooley 2015). Defenders of democracy should be just as vigilant.

ACKNOWLEDGMENTS

Thanks for helpful comments from Michael Joseph, Robert Carroll, Jennifer Gandhi, Zhaotian Luo, Milan Svolik, Stephen Hanson, Adam Przeworski, Grigore Pop-Eleches, Beatriz Magaloni, Daniel Treisman, audience members at American Political Science Association 2017, Midwest Political Science Association 2018, and Yale University, as well as three anonymous reviewers and Editors Sean Gailmard and Tom Clark.

REFERENCES

Acemoglu, Daron, and James A. Robinson. 2006. *Economic Origins of Dictatorship and Democracy*. Cambridge: Cambridge University Press.

Ahlquist, John S., Nahomi Ichino, Jason Wittenberg, and Daniel Ziblatt. 2018. "How Do Voters Perceive Changes to the Rules of the Game? Evidence from the 2014 Hungarian Elections." *Journal of Comparative Economics* 46 (4): 906–19.

Bayer, Markus, Felix S. Bethke, and Daniel Lambach. 2016. "The Democratic Dividend of Nonviolent Resistance." *Journal of Peace Research* 53 (6): 758–71.

BBC. 2012. "Ghana Election: NPP Considers Challenge to John Mahama Win." BBC News, December 10. http://www.bbc.com/news/world-africa-20671298.

Beissinger, Mark R. 2013. "The Semblance of Democratic Revolution: Coalitions in Ukraine's Orange Revolution." *American Political Science Review* 107 (3): 574–92.

Bermeo, Nancy. 2016. "On Democratic Backsliding." *Journal of Democracy* 27 (1): 5–19.

Bitar, Sergio, and Abraham F. Lowenthal, eds. 2015. *Democratic Transitions: Conversations with World Leaders*. Baltimore: Johns Hopkins University Press.

Bohlken, Anjali T. 2010. "Coups, Elections and the Predatory State." *Journal of Theoretical Politics* 22 (2): 169–215.

Boix, Carles, Michael K. Miller, and Sebastian Rosato. 2013. "A Complete Data Set of Political Regimes, 1800–2007." *Comparative Political Studies* 46 (12): 1523–54.

Brancati, Dawn. 2016. *Democracy Protests: Origins, Features, and Significance*. Cambridge: Cambridge University Press.

Cantoni, Davide, David Y. Yang, Noam Yuchtman, and Y. Jane Zhang. 2019. "Protests as Strategic Games: Experimental Evidence from Hong Kong's Antiauthoritarian Movement." *Quarterly Journal of Economics* 134 (2): 1021–77.

Chernykh, Svitlana, and Milan W. Svolik. 2015. "Third-Party Actors and the Success of Democracy: How Electoral Commissions, Courts, and Observers Shape Incentives for Electoral Manipulation and Post-election Protests." *Journal of Politics* 77 (2): 407–20.

Cooley, Alexander. 2015. "Countering Democratic Norms." *Journal of Democracy* 26 (3): 49–63.

Coppedge, Michael. 2017. "Eroding Regimes: What, Where, and When?" Working paper.

Dahl, Robert A. 1971. *Polyarchy: Participation and Opposition*. New Haven, CT: Yale University Press.

Enikolopov, Ruben, Alexey Makarin, Maria Petrova, and Leonid Polishchuk. 2017. "Social Image, Networks, and Protest Participation." Working paper.

Fearon, James D. 2011. "Self-Enforcing Democracy." *Quarterly Journal of Economics* 126:1661–708.

Goldsmith, Arthur A. 2008. "Making the World Safe for Partial Democracy? Questioning the Premises of Democracy Promotion." *International Security* 33 (2): 120–47.

Graham, Benjamin A. T., Michael K. Miller, and Kaare Strøm. 2017. "Safeguarding Democracy: Powersharing and Democratic Survival." *American Political Science Review* 111 (4): 686–704.

Higley, John, and Michael Burton. 2006. *Elite Foundations of Liberal Democracy*. Oxford: Rowman & Littlefield.

Houle, Christian. 2009. "Inequality and Democracy: Why Inequality Harms Consolidation but Does Not Affect Democratization." *World Politics* 61 (4): 589–622.

Huq, Aziz, and Tom Ginsburg. 2018. "How to Lose a Constitutional Democracy." *UCLA Law Review* 65 (1): 78–169.

Kosterina, Svetlana. 2017. "Ambition, Personalist Regimes, and Control of Authoritarian Leaders." *Journal of Theoretical Politics* 29 (2): 167–90.

Lijphart, Arend. 2012. *Patterns of Democracy: Government Forms and Performance in Thirty-Six Countries*. 2nd ed. New Haven, CT: Yale University Press.

Linz, Juan J., and Alfred Stepan. 1996. *Problems of Democratic Transition and Consolidation: Southern Europe, South America, and Post-Communist Europe*. Baltimore: Johns Hopkins University Press.

Little, Andrew T., Joshua A. Tucker, and Tom LaGatta. 2015. "Elections, Protest, and Alternation of Power." *Journal of Politics* 77 (4): 1142–56.

Maeda, Ko. 2010. "Two Modes of Democratic Breakdown: A Competing Risks Analysis of Democratic Durability." *Journal of Politics* 72 (4): 1129–43.

Magaloni, Beatriz. 2010. "The Game of Electoral Fraud and the Ousting of Authoritarian Rule." *American Journal of Political Science* 54 (3): 751–65.

Mainwaring, Scott, and Anibal Pérez-Liñnán. 2013. *Democracies and Dictatorships in Latin America: Emergence, Survival, and Fall*. New York: Cambridge University Press.

Marshall, Monty G., and Keith Jaggers. 2017. *Polity IV Project: Political Regime Characteristics and Transitions, 1800–2016*. College Park, MD: Center for Systemic Peace.

Marsteintredet, Leiv, Mariana Llanos, and Detlef Nolte. 2013. "Paraguay and the Politics of Impeachment." *Journal of Democracy* 24 (4): 110–23.

Pérez-Liñnán, Anibal. 2007. *Presidential Impeachment and the New Political Instability in Latin America*. Cambridge: Cambridge University Press.

Powell, Robert. 2006. "War as a Commitment Problem." *International Organization* 60 (1): 169–203.

Przeworski, Adam. 1991. *Democracy and the Market: Political and Economic Reforms in Eastern Europe and Latin America*. Cambridge: Cambridge University Press.

Przeworski, Adam. 2005. "Democracy as Equilibrium." *Public Choice* 123:253–73.

Przeworski, Adam, Gonzalo Rivero, and Tianyang Xi. 2015. "Elections as a Conflict Processing Mechanism." *European Journal of Political Economy* 39:235–48.

Ritter, Zacc, and Jeffrey M. Jones. 2018. "Media Seen as Key to Democracy but Not Supporting It Well." Gallup, January 16. http://news.gallup.com/poll/225470/media-seen-key-democracy-not-supporting.aspx.

Roeder, Philip G., and Donald S. Rothchild, eds. 2005. *Sustainable Peace: Power and Democracy after Civil Wars*. Ithaca, NY: Cornell University Press.

Santora, Marc, and Helene Bienvenu. 2018. "Secure in Hungary, Orban Readies for Battle with Brussels." *New York Times*, May 11.

Schedler, Andreas. 1998. "What Is Democratic Consolidation?" *Journal of Democracy* 9:91–107.

Seawright, Jason. 2012. *Party-System Collapse: The Roots of Crisis in Peru and Venezuela*. Stanford, CA: Stanford University Press.

Simon, Joel. 2006. "Muzzling the Media: How the New Autocrats Threaten Press Freedoms." *World Policy Journal* 23 (2): 51–61.

Sparks, Allister. 1996. *Tomorrow Is Another Country: The Inside Story of South Africa's Road to Change*. Chicago: University of Chicago Press.

Svolik, Milan. 2015. "Which Democracies Will Last? Coups, Incumbent Takeovers, and the Dynamic of Democratic Consolidation." *British Journal of Political Science* 45 (4): 715–38.

Svolik, Milan. 2017. "When Polarization Trumps Civic Virtue: Partisan Conflict and the Subversion of Democracy by Incumbents." Working paper.

Waldner, David, and Ellen Lust. 2018. "Unwelcome Change: Coming to Terms with Democratic Backsliding." *Annual Review of Political Science* 21:93–113.

Weingast, Barry R. 1997. "The Political Foundations of Democracy and the Rule of Law." *American Political Science Review* 91 (2): 245–63.